FLARE-2: In-Context Learning of Attack Classes from a Few Labelled Flows Across Network Datasets

Quang-Vinh Dang
British University Vietnam, Hung Yen, Vietnam
vinh.dq4@buv.edu.vn ORCID 0000-0002-3877-8024

Abstract

Flow-based intrusion detectors that score above 0.99 ROC-AUC inside one dataset often fall to chance on another. On eight public flow datasets from six laboratories, we find that this is not a tuning problem: no flow feature keeps the same direction of association with the attack label across datasets, and a classifier trained on the pooled other datasets reaches only 0.32 to 0.95 ROC-AUC on a held-out one. Domain-adaptation methods that assume a fixed $P(y \mid x)$ cannot fix this. We therefore ask what other networks can still teach a new network when only $k = 4$ to 128 of its flows are labelled. FLARE-2 is a transformer meta-trained on episodes drawn from the other datasets. It reads the detection rule from the labelled flows given at test time instead of storing a rule, so no model is trained or tuned on the target. Each episode uses a random label space (binary, one class against benign, class against class, or N-way), random class-to-slot assignments, per-episode rank normalisation, and random feature sign flips; the model has no feature identity and a class-equivariant matching readout. Checkpoints are selected on a held-out source dataset. Over the eight targets, FLARE-2 is statistically indistinguishable from random forest and CatBoost on binary ROC-AUC (mean difference to target-only CatBoost -0.001, to random forest $+0.004$), and clearly better than logistic regression and kNN. At attack-class level it has higher per-class ROC-AUC than random forest and CatBoost on 7 of 8 targets and a lower false-positive rate, at lower recall at the default threshold. We report where it does not help: easy targets and $k \geq 32$. Code, configurations and all raw per-run results are public.

1 Introduction

Machine-learning intrusion detection on network flows is usually evaluated by splitting one dataset into train and test. Under that protocol, gradient-boosted trees reach ROC-AUC above 0.99 on CIC-IDS2017, USB-IDS-1 and similar sets. Studies that move the same model to a second dataset report a collapse [2, 5, 8, 17], and a line of work asks whether domain adaptation helps [18]. An earlier study of ours trained on one dataset, measured the distance of a new dataset to it with an isolation-forest score, and fine-tuned CatBoost on labelled target rows [7]. That approach needs target labels, and we show below that source pre-training does not even beat training on the target labels alone.

This paper builds on the diagnosis and proposes a different use of source data. We make three contributions.

1. **Evidence that cross-dataset failure is concept shift.** On eight datasets from six laboratories and three flow extractors, no common flow feature keeps its sign of association with the attack label (0 of 14), a domain classifier recognises the dataset of a flow with 91.5% accuracy, and a pooled-source CatBoost reaches 0.32–0.95 ROC-AUC on the held-out dataset.

2. **FLARE-2**, a few-label meta-learner for flow data. It is permutation-invariant over features, accepts any feature set, and handles binary detection and attack-type classification with the same readout.

3. **A paired evaluation at the level of the target dataset and the attack class**, with exact Wilcoxon tests, bootstrap intervals over targets and a candid account of the cases FLARE-2 loses.

We do not claim a large gain in binary ROC-AUC over tuned tree ensembles. The result is parity with fixed-setting tree ensembles without any per-target fitting of the network, a gain in per-class ranking, and one model family for binary and attack-type output.

2 Related work

Cross-dataset intrusion detection. Layeghy and Portmann [17] train on one NetFlow-v2 dataset and test on another and report attack-class F1 below 1% for several pairs; Cantone et al. [5] report MCC around 0.3 between CIC-IDS2017 and CIC-IDS2018; D'hooge et al. [8, 9] study generalisation per attack class; Apruzzese et al. [2] cross-evaluate detectors across datasets and Engelen et al. [11] document labelling errors in CIC-IDS2017; Arp et al. [4] and Apruzzese et al. [3] discuss pitfalls of ML evaluation in security. Sarhan et al. [23] propose standard NetFlow feature sets to ease cross-dataset use. Domain-invariant training [18], built on adversarial alignment [13] or covariance alignment [26], assumes the labelling function is shared. Our concept-shift controls test that assumption directly.

Few-shot and in-context learning. Matching and prototypical networks [25, 27] and gradient-based meta-learning [12] learn from episodes. Prior-fitted networks [21] and TabPFN [16] fit a transformer to synthetic tabular tasks and classify by attention to the labelled rows. FLARE-2 follows this in-context recipe but meta-trains on *real* tasks from other networks, adds a task distribution designed for the cross-network case, and uses a matching readout that fits a variable label space. Set-based architectures [19] motivate our permutation invariance over rows and features.

3 Problem, data and evidence of concept shift

3.1 Datasets and common flow schema

Table 1 lists the eight datasets. CIC-IDS2017 and CIC-IDS2018 share a laboratory and extractor, so we report the benchmark across six laboratories and three extractors (CICFlowMeter, Argus, Zeek). Extractors export different features, so we keep only features of the same physical meaning and units: five base features (duration, forward and backward packets, forward and backward bytes) plus nine derived rates and ratios, all passed through $\log(1 + x)$. CTU-13 supports only a four-feature minimal tier; FLARE-2 takes it as a target without change because it accepts any number of features. Ports and protocol are excluded. Rows are sampled by a seeded 64-bit hash of the harmonised features and label (a bottom-k sketch), which gives a uniform sample of distinct rows and preserves the natural attack prior.

3.2 Setting

We use leave-one-dataset-out (LODO): the target dataset is never seen in meta-training or checkpoint selection. At test time a method receives a support set of $k \in \{4, 8, 16, 32, 64, 128\}$ labelled target

Table 1: Datasets. Extractor: CIC = CICFlowMeter. "Classes" counts attack classes with enough test rows for class-level evaluation.

Dataset	Laboratory	Extractor	Reference
CIC-IDS2017	UNB CIC	CIC	[24]
CIC-IDS2018	UNB CIC / CSE	CIC	[24]
USB-IDS-1	Univ. of Sannio	CIC	[6]
InSDN	Univ. College Dublin	CIC	[10]
UNSW-NB15	UNSW Canberra	Argus	[20]
ToN-IoT	UNSW Canberra	Zeek	[1]
IoT-23	Stratosphere Lab, CTU	Zeek	[15]
CTU-13	Stratosphere Lab, CTU	Argus	[14]

Table 2: Zero-shot ROC-AUC on the target test split ($k = 0$). LODO: CatBoost on all other datasets pooled. Pairwise: mean and best single source from the pairwise benchmark (not computed for CTU-13). Full label: CatBoost trained on the full target train split.

Target	CatBoost LODO	pairwise mean	pairwise best	full label
CIC-IDS2017	0.832	0.559	0.761	0.999
CIC-IDS2018	0.656	0.547	0.698	0.968
CTU-13	0.358	–	–	0.962
InSDN	0.730	0.598	0.827	0.998
IoT-23	0.322	0.370	0.460	0.950
ToN-IoT	0.608	0.593	0.942	0.999
UNSW-NB15	0.485	0.585	0.781	0.977
USB-IDS-1	0.952	0.695	0.965	1.000

flows with at least one flow per class, and may use no other target label. Support sets are drawn 10 times per (target, k) with fixed seeds and are identical for every method. Evaluation is on the target test split. For class level we use $k \in \{2, 4, 8, 16\}$ labelled flows *per class* and 5 draws.

3.3 Why transfer fails: three controls

No stable feature direction. For each of the 14 common features we compute the ROC-AUC against the attack label inside every dataset. None has AUC above 0.5 in all datasets or below 0.5 in all datasets (0 of 14 consistent; e.g. `total_bytes` ranges from 0.025 to 0.760).

Datasets are identifiable. A CatBoost domain classifier recognises the source dataset of a flow with 91.5% accuracy on seven datasets (chance 14.3%), 91.8% on benign flows alone. A model given unlabelled context can therefore recognise which dataset it is in and apply a memorised rule; our first attempt with unlabelled context rows failed for this reason (Sect. 4).

Source pre-training does not help the target. Over 42 source–target pairs, a source-pretrained CatBoost fine-tuned on 0.1% of target labels reaches 0.835 ROC-AUC, below the 0.928 of CatBoost trained on those labels alone; with 300 or 500 extra trees the gap narrows (0.911 vs 0.932) but does not close. Table 2 shows the zero-shot picture: a CatBoost trained on all other datasets (LODO) scores from 0.322 (IoT-23) to 0.952 (USB-IDS-1), below chance on three targets (IoT-23, CTU-13, UNSW-NB15).

4 Method

FLARE-2 classifies a query flow by attending to labelled support flows of the same episode. At test time the episode is the k support flows and the query flows; nothing is trained.

Episode normalisation. Within an episode, each feature is mapped to its Gaussian rank over support and query rows and clipped to $[-4, 4]$. Absolute scales, which differ across extractors, therefore never reach the network. Missing features are absent tokens.

Tokens and layers. Flow i with features $x_{i1}, \ldots, x_{id}$ becomes d feature tokens $t_{ij} = \mathrm{MLP}_v(x_{ij})$ with a shared scalar embedding, plus one label token. There is no feature-identity embedding, so the model cannot memorise that "feature 7 high means attack" and is invariant to feature order. Six layers of width 128 with 4 heads alternate *feature attention* (tokens of one row attend to each other) and *row attention* (rows attend across the episode). Support rows attend to support rows; a query row attends to support rows and itself only, so predictions do not depend on which other queries share the batch.

Label slots and matching readout. Class identities differ between episodes, so each support class is given a random one of $M = 16$ label-slot embeddings as input. The readout does not index slots. Each row is summarised by the layer-normalised mean r_i of its tokens; a query gives $q = W_q r$, a support row s gives key $k_s = W_k r_s$, and the logit of an active class j with support S_j is

$$\ell_j = \log \sum_{s \in S_j} \exp\Big(\frac{\langle q, k_s \rangle}{\sqrt{w}\,\tau}\Big) - b \log |S_j|, \tag{1}$$

with $w = 128$, a learned temperature $\tau > 0$ and a learned $b \in [0, 1]$ that interpolates between sum and mean pooling over a class's keys. A softmax over the N active classes is trained with cross-entropy. The head is equivariant to permutations of classes and slots by construction and serves binary detection ($N = 2$) and attack-type classification alike. A slot-indexed linear head plateaued in training and was replaced by this one. At test time we average class logits over 8 members with different sign flips and class-to-slot maps.

Meta-training episodes. Each episode draws a source dataset (weighted by $\sqrt{\text{size}}$) and a *task*: binary attack vs benign (probability 0.25), one attack class vs benign (0.25), class vs class (0.15), or N-way with $N \sim U\{3..6\}$ among benign and attack classes (0.35). Raw attack types with fewer than 30 rows are dropped. Support size is log-uniform on $\{4, \ldots, 128\}$ with at least one row per class; the query attack prior is drawn from $U(0.05, 0.95)$ so no fixed prior can be learned; 30% of episodes combine benign rows of one dataset with attack classes of another; 2% of support labels are flipped; each feature is sign-flipped independently with probability 0.5. A monotone warp was also tried; it has no effect under rank normalisation and was dropped. Because both the meaning of a class and the sign of every feature change from episode to episode, the only reliable strategy is to read the rule from the support set. This is the direct response to the concept shift of Sect. 3.3.

Held-out-source model selection. Our first variant added unlabelled context rows from the target network. It reached 0.97 AUC on validation episodes from the meta-training datasets but fell to 0.66–0.76 on the unseen target: with datasets identifiable at 91%, six binary tasks are easy to memorise. In FLARE-2, one source dataset (rotating, never the target) is excluded from training

and used only to select the checkpoint with the lowest validation loss. The unlabelled-context variant is not part of FLARE-2.

Training. AdamW (learning rate 3×10^{-4}, weight decay 0.01, cosine schedule, 2k warm-up steps), 20k steps, 8 episodes per step, 256 query rows, gradient clipping at 1.0, fp16 autocast on a T4 GPU. One model is trained per target (LODO), about 1 hour each. The label-slot readout computes in float32 to avoid half-precision overflow.

5 Experimental protocol

Baselines on identical supports. Random forest (RF), CatBoost [22], logistic regression (LR) and kNN trained on the k target labels only; CatBoost pretrained on the sources and fine-tuned on the k labels (the earlier approach); and, as zero-shot references, CatBoost trained on all other datasets and the pairwise single-source results. Baselines use fixed standard settings and no tuning (RF: 300 trees; CatBoost: 300 iterations, depth 6, learning rate 0.05; LR and 5-NN on standardised features; class-balanced weights), since selecting hyperparameters from a handful of labels is itself unreliable; a baseline tuned with extra target labels could be stronger. For N-way, baselines are the multi-class versions trained on the labelled k per class.

Metrics and statistics. ROC-AUC (binary and per class), recall and false-positive rate at threshold 0.5, and macro-F1 for N-way classification. Draws of one target are not independent, so the unit of analysis is the *target*: the mean difference of FLARE-2 minus baseline over draws, then an exact two-sided Wilcoxon signed-rank test over the 8 targets (smallest attainable $p = 0.0078$), Holm-adjusted over the five baselines, and a 95% bootstrap interval over targets. For class-level results we also report units of (target, class), with a warning that classes of one target are correlated; the target-level test is the conservative one.

6 Results

6.1 Binary detection

Table 3 and Fig. 1 show ROC-AUC versus k. With $k = 4$ labelled flows, FLARE-2 already beats the zero-shot pooled CatBoost on five of eight targets (CTU-13, IoT-23, ToN-IoT, UNSW-NB15, USB-IDS-1; Tables 2, 3); on the other three (CIC-IDS2017, CIC-IDS2018, InSDN), where pooled sources transfer well, 16 to 32 labelled flows are needed to catch up. Table 4 gives the paired statistics. Against the strongest baselines the difference is small and not significant: mean over k, FLARE-2 minus target-only CatBoost is -0.001 and minus RF is $+0.004$; both intervals contain zero and the win/loss counts over targets are 4/4 and 5/3. Against LR and kNN the gain is large: FLARE-2 is higher on all 8 targets for the mean over k ($+0.075$ and $+0.110$, $p_{\text{Holm}} = 0.039$ for both). Against source-pretrained CatBoost fine-tuned on the labels, FLARE-2 is better for $k \geq 32$ (7 of 8 targets, $p_{\text{Holm}} = 0.047$) and not different below.

Where FLARE-2 helps and where it does not is target-dependent (Fig. 2). It is ahead of RF at $k = 4$ on CIC-IDS2017 ($+0.067$) and ahead at $k = 8$–16 on InSDN and ToN-IoT, but behind at $k = 4$ on InSDN, IoT-23 and UNSW-NB15 and behind at $k = 16$ on USB-IDS-1 (-0.036) and CIC-IDS2017 (-0.027). Targets whose attacks are easy (USB-IDS-1, where pooled zero-shot CatBoost already reaches 0.952) leave little to gain.

Table 3: Binary ROC-AUC, mean over 10 support draws. F2: FLARE-2; RF: random forest; CB: CatBoost on target labels; CB-ft: source-pretrained CatBoost fine-tuned on the labels. Bold marks the best of the four.

	$k = 4$				$k = 16$				$k = 64$			
Target	F2	RF	CB	CB-ft	F2	RF	CB	CB-ft	F2	RF	CB	CB-ft
CIC-IDS2017	0.785	0.718	0.754	**0.841**	0.861	**0.888**	0.861	0.880	0.965	0.966	**0.969**	0.956
CIC-IDS2018	0.605	0.599	0.604	**0.691**	0.695	0.657	0.711	**0.733**	**0.892**	0.884	0.880	0.875
CTU-13	**0.599**	0.599	0.599	0.470	0.704	0.700	**0.718**	0.648	0.797	**0.806**	0.797	0.783
InSDN	0.644	0.671	0.690	**0.804**	**0.901**	0.872	0.875	0.890	**0.960**	0.956	0.954	0.949
IoT-23	0.785	0.825	**0.839**	0.763	**0.877**	0.876	0.852	0.857	**0.905**	0.894	0.900	0.891
ToN-IoT	0.720	0.716	**0.723**	0.679	**0.899**	0.861	0.885	0.893	0.961	0.968	**0.968**	0.960
UNSW-NB15	0.647	0.679	**0.682**	0.625	0.794	**0.809**	0.788	0.767	0.870	**0.875**	0.871	0.859
USB-IDS-1	0.978	0.958	0.972	**0.983**	0.957	0.993	**0.997**	0.994	0.997	0.999	**0.999**	0.998

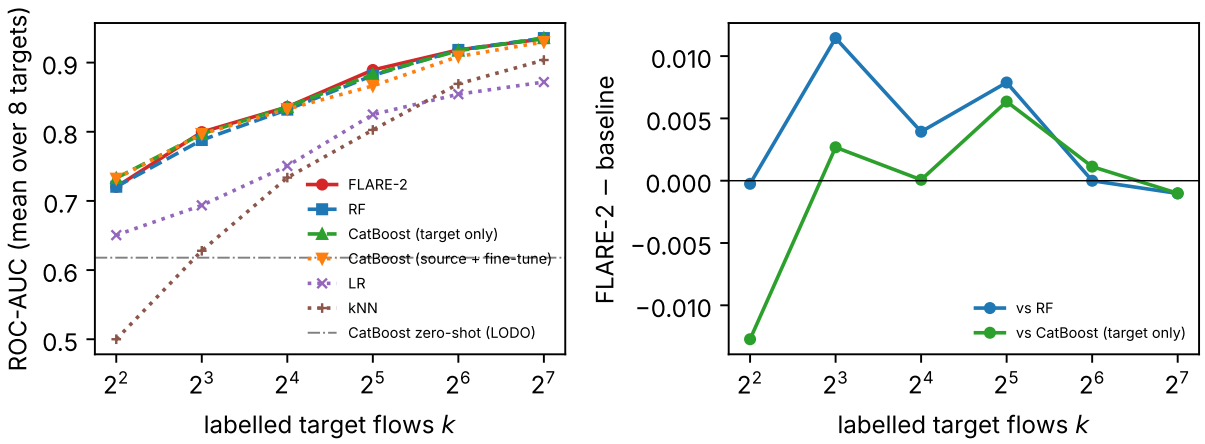


Figure 1: Left: ROC-AUC against k, mean over the eight targets. Right: FLARE-2 minus RF and minus target-only CatBoost.

6.2 Attack-class level

Table 5 summarises per-class detection (class c against benign, k labelled flows of each) and N-way attack-type classification over 57 (target, class) detection tasks. FLARE-2 has higher per-class ROC-AUC than RF and CatBoost on 7 of 8 targets (mean +0.014 and +0.015 over targets, $p = 0.016$ before correction; the unit-level $p < 0.001$ is optimistic because classes within a target are correlated). It has a lower false-positive rate (by 0.05) and a lower recall (by 0.03 to 0.04) at the default threshold of 0.5, so part of the ROC-AUC gain is a different operating point and not a free gain in detection. On N-way macro-F1 FLARE-2 is higher than RF and CatBoost on average (+0.026, +0.022) but on only 3 of 8 targets, so the difference is not significant; it is significantly higher than LR and kNN. Per-target values are in Table 6.

6.3 Ablations

Ablations are running; this section is completed when they finish.

7 Limitations

- One meta-training seed and one set of 10 support draws per target; the confidence intervals bound the target-to-target spread, not the training variance.

Table 4: Paired differences in binary ROC-AUC (FLARE-2 minus baseline), unit = target (8). W/L: targets where FLARE-2 is higher / lower. p_{Holm}: exact Wilcoxon over targets, Holm-adjusted over baselines.

Scope	Baseline	ΔROC-AUC	95% CI	W/L	p_{Holm}
mean over k	RF	+0.004	[-0.003, +0.010]	5/3	1.000
mean over k	CatBoost (target)	-0.001	[-0.004, +0.003]	4/4	1.000
mean over k	CatBoost (fine-tune)	+0.005	[-0.010, +0.022]	4/4	1.000
mean over k	LR	+0.075	[+0.044, +0.106]	8/0	0.039
mean over k	kNN	+0.110	[+0.085, +0.133]	8/0	0.039
k≤16	RF	+0.005	[-0.004, +0.014]	4/4	1.000
k≤16	CatBoost (target)	-0.003	[-0.009, +0.002]	4/4	1.000
k≤16	CatBoost (fine-tune)	-0.002	[-0.033, +0.030]	4/4	1.000
k≤16	LR	+0.087	[+0.047, +0.128]	8/0	0.039
k≤16	kNN	+0.165	[+0.121, +0.207]	8/0	0.039
k≥32	RF	+0.002	[-0.003, +0.008]	4/4	1.000
k≥32	CatBoost (target)	+0.002	[-0.002, +0.007]	4/4	1.000
k≥32	CatBoost (fine-tune)	+0.012	[+0.006, +0.019]	7/1	0.047
k≥32	LR	+0.064	[+0.034, +0.096]	8/0	0.039
k≥32	kNN	+0.056	[+0.040, +0.073]	8/0	0.039

- FLARE-2 is not better than fixed-setting tree ensembles on binary ROC-AUC; its gains are class-level and conditional on the operating point.
- Eight datasets from six laboratories, with the CIC pair counted separately; results on other networks may differ.
- A training run assigns a very large negative logit to rare queries whose class has no support (label noise at $k = 1$); an optional loss-mask variant exists and its effect is reported in the ablations.
- Flows only; payload features and temporal context are not used.
- The datasets are public laboratory captures with their own labelling errors [11].

8 Conclusion

Concept shift, not tuning, limits cross-dataset flow classification, so a model that carries a decision function across networks is the wrong target. FLARE-2 carries the skill of inferring a rule from a few labelled flows instead. It matches fixed-setting tree ensembles on binary detection without fitting the network to the target, improves per-class ranking, and outputs attack types through one readout. The open problem is the remaining gap to full-label training (0.95–1.00 ROC-AUC) at small k, where better label-efficient queries and multi-seed meta-training are the next step.

Reproducibility. The package, configurations, sampled-row hashes and every raw per-run CSV are in `https://github.com/vinhqdang/cross_ids`; tables and figures are generated by `scripts/make_paper_tables.py` and statistics by `scripts/stats_flare.py`.

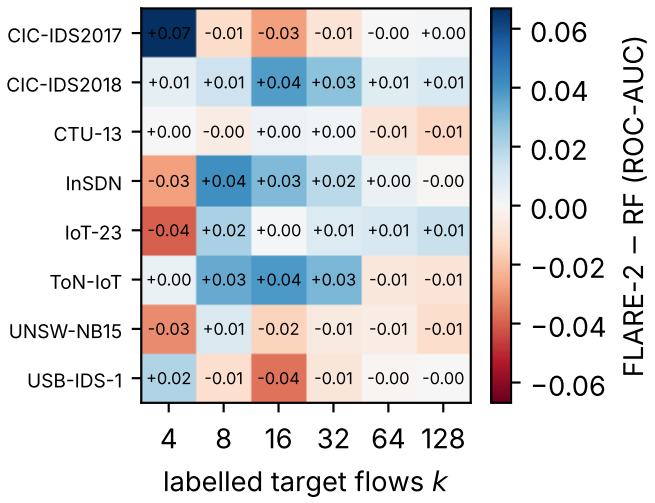


Figure 2: ROC-AUC difference FLARE-2 minus RF per target and k.

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Table 5: Class-level differences, FLARE-2 minus baseline. "targets": mean over the 8 targets of the target mean, count of targets where FLARE-2 is higher, exact Wilcoxon p over targets (uncorrected). "units": mean over (target, class) pairs, $k \in \{2, 4, 8, 16\}$ labelled flows per class. For FPR a negative difference favours FLARE-2.

Metric	Baseline	Δ (targets)	FLARE-2 higher	p	Δ (units)	units
Per-class ROC-AUC	RF	+0.014	7/8	0.016	+0.012	57
Per-class ROC-AUC	CatBoost	+0.015	7/8	0.016	+0.013	57
Per-class ROC-AUC	LR	+0.082	8/8	0.008	+0.075	57
Per-class ROC-AUC	kNN	+0.175	8/8	0.008	+0.167	57
Per-class recall@0.5	RF	-0.041	0/8	0.008	-0.044	57
Per-class recall@0.5	CatBoost	-0.026	1/8	0.016	-0.030	57
Per-class recall@0.5	LR	-0.025	2/8	0.078	-0.030	57
Per-class recall@0.5	kNN	-0.048	1/8	0.023	-0.056	57
Per-class FPR@0.5	RF	-0.053	1/8	0.078	-0.052	57
Per-class FPR@0.5	CatBoost	-0.052	1/8	0.078	-0.050	57
Per-class FPR@0.5	LR	-0.077	1/8	0.016	-0.074	57
Per-class FPR@0.5	kNN	-0.334	0/8	0.008	-0.330	57
N-way macro-F1	RF	+0.026	3/8	0.742	+0.026	8
N-way macro-F1	CatBoost	+0.022	3/8	0.742	+0.022	8
N-way macro-F1	LR	+0.074	7/8	0.016	+0.074	8
N-way macro-F1	kNN	+0.127	8/8	0.008	+0.127	8

Table 6: Per-target class-level results. Left: per-class ROC-AUC at $k = 4$ per class. Right: N-way macro-F1 at $k = 16$ per class.

	Per-class ROC-AUC, k=4			N-way macro-F1, k=16		
Target	F2	RF	CB	F2	RF	CB
CIC-IDS2017	**0.957**	0.942	0.949	**0.504**	0.456	0.448
CIC-IDS2018	**0.889**	0.879	0.875	**0.658**	0.517	0.582
CTU-13	**0.863**	0.853	0.852	**0.299**	0.329	0.350
IoT-23	**0.879**	0.842	0.841	**0.644**	0.562	0.589
ToN-IoT	**0.927**	0.906	0.925	**0.680**	0.705	0.688
UNSW-NB15	**0.857**	0.845	0.839	**0.294**	0.337	0.323
USB-IDS-1	**0.995**	0.998	0.999	**0.461**	0.461	0.468

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